

Code Explanation

Sample Data Generator Code Explanation

Overview

This code is a sample data generator for testing the E2E Policy Services data extraction pipeline. It creates sample data using PySpark and returns a dictionary containing sample DataFrames.

Step 1: Importing necessary libraries and defining the module docstring

The code begins by defining a docstring that describes the purpose of the module. It then imports the necessary libraries, including PySpark and datetime.

```
"""
Sample data generator for testing the E2E Policy Services data extraction pipeline.
"""
from pyspark.sql import SparkSession
from pyspark.sql.types import StructType, StructField, StringType, IntegerType, TimestampType, DecimalType
from datetime import datetime, timedelta
import random
```

Step 2: Defining the create_sample_data function

The `create\_sample\_data` function takes a `SparkSession` object as an argument and returns a dictionary containing sample DataFrames.

```
def create_sample_data(spark: SparkSession):
    """
    Create sample data for testing the ETL pipeline.

    Args:
        spark: SparkSession

    Returns:
        dict: Dictionary containing sample DataFrames
    """
```

Step 3: Creating the schema for T_TX_REQUEST_POLICY DataFrame

The code defines a schema for the `T\_TX\_REQUEST\_POLICY` DataFrame using PySpark's `StructType` and `StructField` classes. The schema includes various fields with different data types.

```
request_policy_schema = StructType([
    StructField("REQUEST_POLICY_ID", IntegerType(), False),
    StructField("REQUEST_ID", IntegerType(), False),
    StructField("TRANSACTION_ID", IntegerType(), False),
    StructField("POLICY_ID", StringType(), False),
    StructField("TRANSACTION_TYPE", StringType(), True),
    StructField("BASE_CD", StringType(), True),
    StructField("SBASE_CD", StringType(), True),
    StructField("PSTA_CD", StringType(), True),
    StructField("PENTT_CD", StringType(), True),
```

```
StructField("ORPOLDT", TimestampType(), True),  
StructField("PUSG_CD", StringType(), True),  
StructField("PUSG_EDT", TimestampType(), True),  
StructField("PUSG_PDT", TimestampType(), True),  
StructField("PWTV_CD", StringType(), True),  
StructField("PWTV2_CD", StringType(), True),  
# ... other fields ...  
1)
```

The rest of the code is not provided, so the explanation will be limited to the given snippet. If you provide the complete code, I can continue the explanation.